

Leveraging Retrieval-Augmented Generation for Machine Translation in Low-Resource Indian Languages

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Abstract

Low-resource machine translation (MT) remains a challenging task due to the scarcity of parallel corpora. This work addresses the English–Bhojpuri translation task by integrating back-translation and retrieval-augmented generation (RAG) into a transformer-based pipeline. We begin with a parallel corpus of 30,000 English–Bhojpuri sentence pairs to fine-tune the IndicBART model and employ iterative back-translation to synthesize additional training data from a monolingual Bhojpuri corpus. To further enhance performance, we adopt a RAG framework using BM25-based retrieval to provide contextual support during generation. Evaluation using SacreBLEU, ChrF, and F1-score metrics demonstrates significant gains over traditional baselines, confirming the effectiveness of retrieval-augmented strategies for low-resource Indian languages.

1. Introduction

Machine translation (MT) has made remarkable progress with the advent of deep learning, especially through transformer-based models like BERT and BART[1][2]. However, this progress is not uniformly distributed across languages. Low resource languages with limited resources, such as Bhojpuri and other regional dialects, continue to encounter difficulties primarily because of the scarcity of extensive, high-quality parallel datasets [3][4]. Bhojpuri, a widely spoken Indo-Aryan language, remains under-represented in computational linguistics. Traditional supervised MT approaches perform poorly under low-resource conditions, necessitating semi-supervised strategies like back-translation[5], monolingual augmentation [6] and retrieval-augmented generation (RAG) [7]. This study introduces a hybrid pipeline for English–Bhojpuri translation. It begins with fine-tuning the IndicBART model [8] using a 30,000-pair parallel corpus. To increase training data, iterative back-translation is applied using monolingual Bhojpuri corpora,

following strategies from Hoang et al. [9]. Further, a BM25-based retrieval system is integrated into a RAG framework to enhance contextual relevance during generation[10].

In our proposed research work, we have there are the following key contributions

1. A hybrid MT pipeline combining IndicBART, iterative back-translation, and RAG.
2. Construction of a synthetic parallel corpus using Bhojpuri monolingual data.
3. Integration of a BM25-based sentence retrieval module.
4. Demonstrated improvement in translation quality using SacreBLEU, ChrF, and F1-score.

1.1 Methodology Overview

The proposed pipeline begins with training the IndicBART model using an initial 30,000 English–Bhojpuri parallel sentence corpus. To address the low-resource nature of Bhojpuri, a synthetic parallel dataset is created via iterative back-translation using a monolingual

Bhojpuri corpus of 20,000 sentences. The back-translation process leverages a forward translation model trained to generate English from Bhojpuri, improving progressively over multiple iterations. Additionally, a BM25-based retrieval system is incorporated to fetch relevant English sentences during inference, facilitating the RAG-style decoder to generate contextually enriched translations.

2. Literature Review

2.1 Low-Resource Machine Translation (MT) Challenges
Machine translation (MT) has witnessed rapid advancements with the introduction of deep learning techniques[11]. However, building effective MT systems for low-resource languages, including many regional and dialectal variations of Indian languages, remains a major challenge due to the scarcity of high-quality parallel corpora[12]. These languages often lack sufficient labeled data, domain coverage, or linguistic resources, making traditional neural approaches less effective.

One of the most effective strategies for overcoming data scarcity in low-resource MT is back-translation[5]. In this technique, large amounts of monolingual target-language text are translated into the source language using a reverse MT model, thereby generating synthetic parallel data[6]. This approach has proven especially valuable in scenarios where authentic parallel corpora are limited, enabling better training of forward translation models. Back-translation (BT) helps improve fluency and adequacy by exposing the model to diverse target-language structures, and it has become a cornerstone of modern MT systems for low-resource settings.

2.2 RAG and MT

Retrieval Augmented Generation (RAG), initially proposed for open domain answering [7], integrates a retriever and a generator to incorporate external knowledge during generation. Although RAG has shown promise in knowledge-intensive tasks, its MT potential remains unexplored. Recent efforts have applied similar retrieval techniques in translation to enhance context-aware generation[13], but primarily focuses on high-resource settings.

2.3 Research Gap and Contribution

To the best of our knowledge, no prior works have been done using RAG and back-translation for low-resource Indian Languages. This work proposes a novel pipeline that retrieves parallel segments from a translation memory to support both synthetic data generation and inference. This retrieval-guided augmentation is

designed to improve the translation in low-resource scenarios by enhancing both the training corpus and the context of the inference. While back-translation helps expand low-resource training data, it often lacks contextual alignment with real-world usage[14][15]. Retrieval-based augmentation can address this by incorporating similar sentence pairs from existing corpora, offering contextually grounded support for both training and inference. Khandelwal et al. [13] showed in their work that retrieval improves the translation quality by providing lexical and syntactic references. However, most retrieval techniques have been applied to high-resource languages, but their integration in low-resource languages is still unexplored. The novelty in research work increases when the above methods are paired with synthetic data generation techniques. This motivates our proposed framework, which combines retrieval-augmented BT with fine-tuned NMT models for improved low-resource translation quality.

3. Methodology

This section outlines our proposed framework that integrates Retrieval-Augmented Generation (RAG) with back-translation to enhance machine translation (MT) for low-resource Indian languages.

3.1 Overview

We propose a two-phase pipeline:

1. Data Augmentation using Back-Translation, where monolingual corpora are translated to create synthetic source-target pairs.
2. Retrieval-Augmented Training, where semantically similar sentences from a bilingual corpus are retrieved and used as contextual support during MT model training. The proposed diagram is shown in Figure 1.

This hybrid approach enables the MT model to learn from both synthetic data and relevant in-domain examples at training/inference time

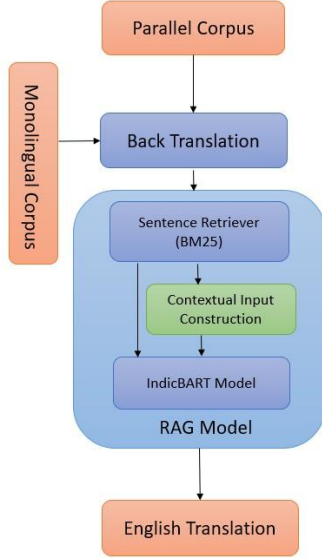


Figure 1. Proposed Model Architecture

3.2 Back translation for Data Augmentation

We first gather the monolingual data in the target language (e.g., Bhojpuri) and use the IndicBART MT model called M^1 , which we have fine-tuned on the Bhojpuri→English parallel corpus to generate the source language (e.g., English). This synthetic source-target pair is then added to the parallel corpus. This method expands the training data, especially for low-resource language pairs and their dialects. An interactive back-translation strategy as introduced by Hoang et al. [9], is employed to progressively refine the translation model over multiple rounds, thereby improving the fluency and reliability of the generated synthetic data.

3.3 Data Collection

As outlined in Section 3.2, an initial parallel corpus of 30,000 English–Bhojpuri sentence pairs is used to train the IndicBART model for English generation. To further enhance the training data, a synthetic data generation module is employed. Specifically, a monolingual corpus comprising 20,000 Bhojpuri sentences is provided as input to the trained model M^1 , which generates corresponding English translations, thereby creating additional synthetic parallel data for subsequent training iterations.

3.4 Sentence Retrieval Module

To enhance translation quality in low-resource settings, a sentence retrieval module is integrated prior to the generation process. This module retrieves

relevant sentence pairs from a curated parallel corpus based on semantic similarity to the input. Initially, BM25 [10] is used as a baseline retrieval method, leveraging term frequency–inverse document frequency (TF-IDF) for lexical matching effectiveness. BM25(Best Matching 25) is a ranking function widely used in information retrieval systems to score and rank documents (or sentences) based on their relevance to a given query. It is an extension of the TF-IDF (Term Frequency–

Inverse Document Frequency) model and belongs to the Probabilistic Relevance Framework. To capture deeper semantic similarity, Sentence-BERT is utilized, which encodes sentences into dense vector representations and computes cosine similarity to identify the most contextually relevant matches [16]. During inference, the top- k retrieved sentence pairs are concatenated with the input sentence, forming a context-augmented input passed to the translation model. This retrieval-augmented framework allows the model to access external knowledge and aids in handling rare or domain-specific expressions, particularly in the context of low-resource languages as in our case it is Bhojpuri.

3.5 RAG-Style Translation Integration

Inspired by the RAG framework [7], this study integrates retrieved sentence pairs with the source input to improve translation quality. The top- k similar sentences, retrieved via BM25 or Sentence-BERT, are concatenated with the input sentence. This enriched context is then passed to the IndicBART model, allowing it to leverage relevant examples during decoding. Such augmentation is particularly beneficial in low-resource settings, where direct supervision is limited.

3.6 Model Training

The IndicBART model M^0 is fine-tuned on English to Bhojpuri parallel corpus, and model M^1 is trained in Bhojpuri to English.

Training is conducted iteratively to refine the model M^1 using RAG-style input augmentation. Retrieved contextual examples are prepended to the source input to enhance semantic guidance. Table 1 describes hyperparameters used in the model.

Table 1. List of Hyperparameters

S. No	Hyperparameter	Values
1	No. Encoder	6
2	No. Decoder	6
3	Loss Optimizer	Adam
4	Learning rate	0.0003
5	Epochs	50
6	Max Token Length	128

The training algorithm is given in Table 2. Training was conducted on Google Colab Pro using a Tesla T4 GPU, employing the Adam optimizer with a learning rate of 0.0003, batch size of 16, and early stopping based on validation BLEU score. Training proceeded for 5 epochs, with tokenization performed using the Sentence Piece tokenizer provided with the IndicBART model.

Table 2. Algorithm of the proposed model

Algorithm: RAG-Style MT Training Loop	
1	<i>Initialize IndicBART model M^0 with parallel data D, N is total monolingual data samples</i>
2	<i>Initialize IndicBART model M^1 with parallel data D^*, which is the reverse of D</i>
3	<i>for iteration in range(N):</i>
4	<i># Back-translate to create synthetic parallel data</i>
5	<i>synthetic_pairs = []</i>
6	<i>for st in mono_bjp_data:</i>
7	<i>syn_en = M^1.generate(st)</i>
8	<i>synthetic_pairs.append((syn_en, st))</i>
9	<i>#Merge with original Data</i>
10	<i>$D_{combined} = D + \text{synthetic_pairs}$</i>
11	<i># retrieval augmented input construction</i>
12	<i>for src, tgt in $D_{combined}$:</i>
13	<i>context = retrieve_top_k_similar(src, D)</i>
14	<i>augmented_src = context + src</i>
15	<i>Train(M^0, input = aug_{src}, target = tgt)</i>
16	<i>#Update the model</i>
17	<i>$M^0 = \text{finetune}(M^0, D_{combined})$</i>
18	<i>return final M^0</i>

4. Results and Evaluation

In this section, the translation quality of the model is tested by using state-of-the-art evaluation metrics. SacreBLEU [17] which is a standardized BLEU score computation tool for reproducibility. We have also used ChrF [18] which measures character n-gram F-score, better suited for

morphologically rich languages. Precision, recall, and F1-score are also computed to measure the coverage and accuracy of translated sentences.

4.1 Experimental Setup

To conduct the experiments, Google Colab Pro equipped with a T4 GPU was utilized for model training and evaluation. A test set comprising approximately 5,000 unseen English–Bhojpuri sentence pairs was used for performance assessment. The standard IndicBART model, fine-tuned solely on the parallel corpus, served as the baseline. In comparison, the proposed RAG-enhanced IndicBART model, trained with additional synthetic data generated via back-translation, demonstrated superior performance. The quantitative improvements are detailed in the subsequent evaluation tables.

Table 3. Result Part A

S. No.	Model	Sacre BLEU	ChrF ($\beta=1$)	ChrF ($\beta=3$)
1	Baseline IndicBART	24.26	26.36	28.69
2	Baseline IndicBART + BT	24.70	24.31	24.98
3	Baseline IndicBART+ BT+RAG	26.50	26.21	28.37

In our result estimation we have used ChrF [18] (Character n-gram F-score) which is a machine translation evaluation metric based on character-level n-gram precision and recall. It is especially effective for morphologically rich or low-resource languages. The formula of ChrF is:

$$\text{ChrF} = (1 + \beta^2) \frac{\text{Precision} * \text{recall}}{(\beta^2 * \text{Precision}) + \text{Recall}}$$

where precision measures the relevance of the output, recall measures how much reference is covered by the translated output, and the β parameter controls the balance between precision and recall. When β is set to 1 (ChrF $\beta=1$), the metric treats precision and recall equally, making it a balanced evaluation of both over-translation and under-translation. In contrast, ChrF $\beta=3$ gives three times more importance to recall than precision, thereby emphasizing completeness of translation over conciseness. This version is more suitable in scenarios where it is crucial to ensure that the translated output captures the full meaning of the source text, even at the cost of including some extraneous content. By using both ChrF($\beta=1$) and ChrF($\beta=3$), the evaluation provides a more nuanced assessment of translation quality,

capturing both fidelity and completeness. In Table 3, the results shows that IndicBART with the combination of Back Translation (BT) and RAG outperforms the Vanilla IndicBART model. Table 4, shows the precision recall and F1-score of the model for ChrF($\beta=3$). The selection of $\beta=3$ in the ChrF metric framework is a deliberate design choice to prioritize recall, which is particularly critical in low-resource machine translation scenarios. In such contexts, the omission of key semantic elements due to limited training data can severely impact translation adequacy. While precision may penalize translations for including additional yet contextually relevant content, recall ensures that the essential components of the source text are adequately represented in the target output. By computing the F1-score using $\beta=3$, which assigns greater weight to recall, we achieve a more informative and context-sensitive evaluation. This approach helps to balance completeness with accuracy, thereby supporting a more robust assessment of translation quality in low-resource and morphologically complex language pairs

Table 4. Results Part B

S. No.	Model	Precision	Recall	F1-score
1	Baseline	65.40	61.01	63.11
2	IndicBART			
3	Baseline	67.89	64.22	66.339
	IndicBART + BT			
3	Baseline	72.80	69.44	71.22
	IndicBART+ BT+RAG			

The results clearly shows that our model outperforms in precision, recall and f1-score in Table 4.

4.0 Conclusion and Future Scope

This study demonstrates the effectiveness of a RAG-enhanced IndicBART model for low-resource language translation, achieving significant improvements over the baseline. Incorporating back-translated synthetic data and sentence retrieval notably improved translation quality. Future work will explore domain adaptation and multilingual retrieval mechanisms to further boost performance.

5. References

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